

1. Question 1
 - a. Yes because the slope confidence interval does not contain zero. The implied level of significance is 0.05.
 - b. Statistically it is fine to have a negative bound because the estimate is just uncertain, not actually that the sales will be negative.
2. Question 2: There is not enough evidence to reject the null hypothesis because of the high p-value, so we can't really determine that spending more on advertising has a linear relationship on units sold.
3. Question 3
 - a. Prediction interval because we are wanting to observe a new value for an individual value in the humidity level.
 - b. Confidence interval because we are focusing on the population mean.
 - c. Prediction interval because we are wanting to predict a specific value in the future, not population mean.
4. Question 4: It is possible that R^2 will not be zero for the complete set of 30 cases. The next 20 cases might fit well linearly and thus R^2 will not be zero. It is also possible that R^2 does not equal zero for the first 10 cases but equals zero for all 30 cases because the 20 cases show no linear relationship to the point where they cancel out the other cases and thus R^2 becomes zero.

$$\begin{aligned}
5.) \quad & E(y_0 - \hat{y}_0) \\
&= E(\beta_0 + \beta_1 x_0 + \varepsilon_0 - \hat{y}_0) \\
&= E(\beta_0 + \beta_1 x_0 + \varepsilon_0 - (\hat{\beta}_0 + \hat{\beta}_1 x_0)) \\
&= E(\beta_0 + \beta_1 x_0 + \varepsilon_0 - \hat{\beta}_0 - \hat{\beta}_1 x_0) \\
&= E(\beta_0) + E(\beta_1 x_0) + \underbrace{E(\varepsilon_0)}_0 - \underbrace{E(\hat{\beta}_0) - E(\hat{\beta}_1 x_0)}_{\text{unbiased}} \\
&= \beta_0 + \beta_1 \bar{x}_0 - \beta_0 - \beta_1 \bar{x}_0 = \boxed{0}
\end{aligned}$$

$$\begin{aligned}
& \text{Var}(y_0 - \hat{y}_0) \\
&= \text{Var}(\beta_0 + \beta_1 x_0 + \varepsilon_0) - \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0) \\
&= \sigma^2 + \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0) \\
&= \sigma^2 + \text{Var}(\hat{\beta}_0) + x_0^2 \text{Var}(\hat{\beta}_1) + 2x_0 \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\
&= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{SSX} \right) + x_0^2 \left(\frac{\sigma^2}{SSX} \right) + 2x_0 \left(-\sigma^2 \frac{\bar{x}}{SSX} \right) \\
&= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{SSX} + \frac{x_0^2}{SSX} - \frac{2\bar{x}x_0}{SSX} \right) \\
&= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2 + x_0^2 - 2\bar{x}x_0}{SSX} \right)
\end{aligned}$$

$$\boxed{= \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{SSX} \right)}$$